

DYNAMO: Dynamic Skill-Tool Evolution for Vision-Language Agents

Anonymous ACL submission

Abstract

Improving vision-language models (VLMs) on visual reasoning typically requires retraining or hand-designed prompts and tools. We present DYNAMO, a training-free framework in which a frozen VLM agent inspects its own correct and incorrect attempts on a small labeled training subset and evolves two complementary capabilities: reusable reasoning skills for cognitive bottlenecks, and executable visual tools—gated by learned applicability—for perceptual ones, accumulating them in a persistent library without weight updates. Across four visual reasoning benchmarks and six VLM backbones, DYNAMO improves direct inference in 21 of 23 settings (avg. +4.8 acc). The same framework extends to the tool-mastery setting on GTA, where mastery profiling lifts step-level tool selection across all tested backbones. Against task-specific RL (VTool-R1, DeepEyes), DYNAMO matches or approaches RL accuracy at a fraction of the compute (60–94% gap recovery on chart/table; 91.1 vs. 90.1 on V*-7B) and combines additively with RL when available.

1 Introduction

Vision-language models (VLMs) have improved rapidly, yet adapting a frozen VLM to a new visual-reasoning task family still typically requires manually curated SFT data or a hand-designed RL pipeline—per task and per backbone. We ask whether the agent can build its own task-family-specific capability set instead, by inspecting its own behaviour on a small training subset and without any weight updates.

We instantiate this idea in DYNAMO, a training-free framework that evolves two complementary capability types from a small labeled training subset and dynamically equips them at inference. **Skills** are structured Markdown SOPs for *cognitive* bottlenecks, where the visual evidence is available but the agent’s reasoning procedure is weak. **Tools** are short Python programs for *perceptual* bottlenecks,



Figure 1: **Per-task adaptation without weight updates or hand-authored prompts.** Per-task VLM adaptation typically requires manually curated SFT data with a hand-designed RL pipeline for every new task family. DYNAMO instead lets a frozen VLM agent evolve its own skill and tool set per task family from a small labeled training subset—without any weight updates or hand-authored task-specific prompts—and dynamically equips the matching set when a case from that family arrives.

where the agent needs a transformed view—a crop, zoom, contrast adjustment, or rerendering—before the evidence becomes legible. On each iteration over a sampled sub-training set, DYNAMO diagnoses both the correct and the incorrect attempts using their images and reasoning traces, proposes multiple candidate skill+tool combinations, and promotes the highest-validation-accuracy candidate into a persistent library. A mastery phase further learns when each tool is safe to invoke, so that promoted capabilities are deployed selectively

rather than indiscriminately.

We evaluate three questions. **(I) Does DYNAMO work across diverse benchmarks and backbones?** On ChartQA, MathVista, HRBench, and V* across six VLM backbones, skill-tool co-evolution improves direct inference in 21 of 23 model–benchmark settings, averaging +4.8 accuracy points. **(II) Does the framework extend to a tool-mastery setting with a curated tool set?** On GTA, mastery profiling lifts step-level tool selection, argument, and instruction-following accuracy across all tested backbones. **(III) Can DYNAMO match task-specific RL across different task families?** Across three task families—chart, table, and high-resolution perception—we compare DYNAMO to a representative RL method on each (VTool-R1 for chart/table; DeepEyes for perception). DYNAMO matches or approaches RL accuracy at a fraction of the compute (60–94% gap recovery on chart/table; 91.1 vs. 90.1 on V*-7B perception), and combines additively with RL when both are available.

Contributions.

- (1) A **problem formulation** that recasts per-task VLM adaptation as capability evolution: a frozen agent builds its own task-family-specific skill and tool set from a small labeled training subset, without any weight updates or hand-authored task-specific prompts, replacing manual SFT data curation and task-specific RL pipelines.
- (2) A **multi-candidate evolution loop with mastery** that diagnoses both correct and incorrect attempts, generates candidate skill+tool combinations, selects the best by validation accuracy, and learns per-tool applicability boundaries.
- (3) **Empirical evidence** across four visual-reasoning benchmarks, six VLM backbones, the GTA tool-mastery benchmark, and three task families (chart, table, perception) each compared against its representative RL method, showing that capability evolution is a viable training-free alternative—and add-on—to task-specific RL.

2 Related Work

Self-improving agents. A growing line of work lets LLM agents improve from experience without gradient updates: Reflexion (Shinn et al., 2023)

keeps a verbal reflection buffer that resets between episodes; Voyager (Wang et al., 2024a) and ExpeL (Zhao et al., 2024) accumulate persistent skill libraries in game and tool-use domains; AutoManual (Zhao et al., 2024) and EvolveR (Wu et al., 2025b) refine instruction manuals or reasoning strategies via self-play; Trace2Skill (?) distills trajectory-local lessons into transferable agent skills. Whether language models can bootstrap reasoning capabilities entirely without external scaffolding remains debated (?), and broader surveys map the design space for personalised LLM-powered agents (?). All these methods operate in text, code, or game domains, never generating executable visual processing nor deciding whether a bottleneck is cognitive or perceptual. DYNAMO extends the persistent-library paradigm into the visual domain with a multimodal diagnostic that text-only frameworks cannot perform.

Tool creation for language models. A complementary line generates new tools on demand: CREATOR (Qian et al., 2023) prompts an LLM to write Python helpers for problems it cannot solve directly; LATM (Cai et al., 2024) and ToolLLM (Qin et al., 2024) scale this to large API collections. These methods generate tools in textual or mathematical domains and do not profile when a generated tool succeeds or fails—a gap that matters for visual tools, where the same operation can help on most images but hurt on specific patterns (e.g. a sharpening filter degrading low-contrast charts). DYNAMO’s mastery phase profiles each tool on a held-out case set to characterise its applicability before promotion.

Visual reasoning with tools. Several recent systems equip VLMs with visual processing tools. VTool-R1 (Wu et al., 2025a), DeepEyes (Zheng et al., 2025), Adaptive-CoF (Zhang et al., 2025), PixelReasoner (Su et al., 2025), and V-Thinker (Qiao et al., 2025) all rely on SFT and/or RL with curated trajectories to teach a VLM when and how to invoke a fixed tool inventory. ZoomEye (Shen et al., 2025) and ReFocus (Fu et al., 2025) are training-free but assume a fixed zoom/search algorithm or visual-editing interface; Lever LM (?) configures in-context example sequences to leverage pretrained VLMs without re-training. DYNAMO instead generates and validates the skills and tools themselves from the agent’s own behaviour on a small training subset, with no weight updates and no predefined tool inventory.

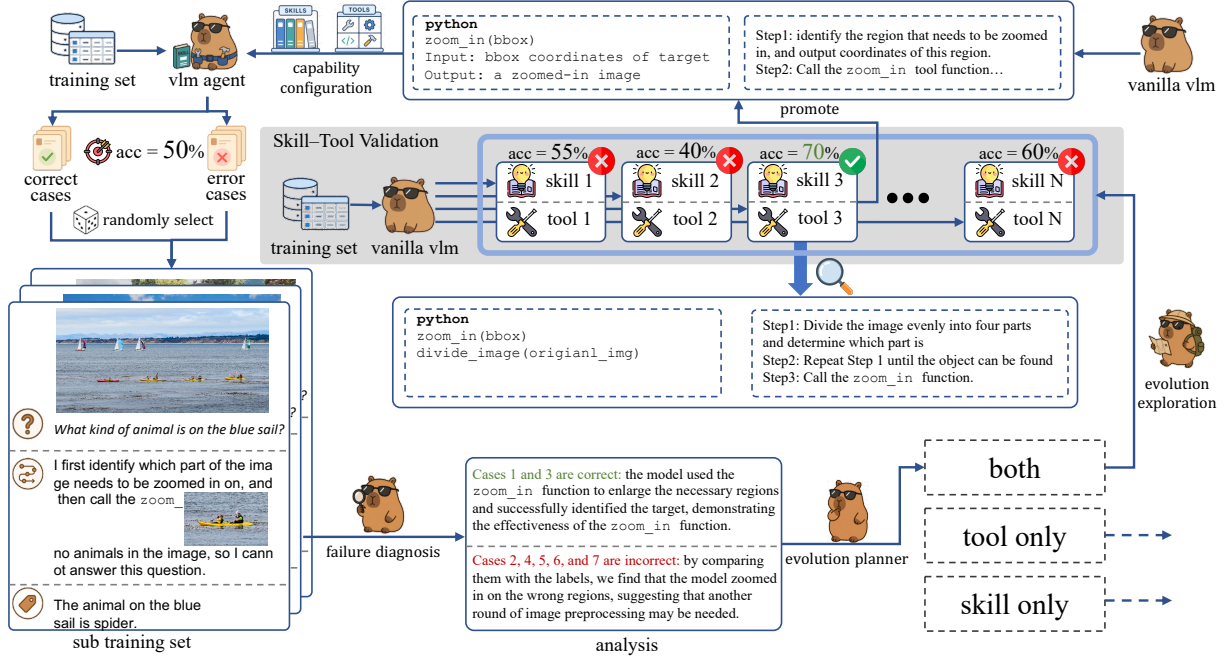


Figure 2: **DYNAMO evolution loop.** On a sampled sub-training set, DYNAMO diagnoses both correct and incorrect attempts, explores candidate skill+tool capabilities, and promotes the best by validation into a persistent library.

Tool-augmented VLMs with predefined tool sets. VisProg (Gupta and Kembhavi, 2023), ViperGPT (Surís et al., 2023), and Chameleon (Lu et al., 2023) compose hand-curated tool libraries (detectors, OCR, arithmetic modules) via program synthesis; MMR-Bench (?) extends evaluation of such systems to multimodal LLM routing across diverse backbones and tools. They handle broad compositional visual tasks effectively but cannot recover when the required tool is missing from the human-designed inventory, and cost-aware alignment frameworks such as EcoAlign (?) highlight the practical pressure to adapt LVLMs efficiently. DYNAMO fills this gap by generating new tools when the current set is insufficient, bridging fixed-tool composition and training-based tool learning.

3 Method

3.1 Problem Formulation

Let $\mathcal{D}_{\text{train}} = \{(x_i, y_i, \mathbf{v}_i)\}_{i=1}^k$ denote a small training subset of k cases, where x_i is the question, y_i the ground-truth answer, and $\mathbf{v}_i$ the associated visual input (one or more images). Let π_θ be a frozen VLM backbone. We define a capability set $\mathcal{C} = (\mathcal{S}, \mathcal{T})$, where $\mathcal{S}$ is a library of *skills* (structured reasoning SOPs) and $\mathcal{T}$ is a library of *tools* (executable Python programs). The agent decomposes solving into a retrieval step and a reasoning

step:

$$f_{\mathcal{C}}(x, \mathbf{v}; \pi_\theta) = \pi_\theta(\cdot \mid x, \mathbf{v}, \text{Retrieve}(x, \mathbf{v}; \mathcal{C})), \quad (1)$$

where Retrieve returns the subset of skills and tools relevant to the current input. Given a held-out validation set $\mathcal{D}_{\text{val}}$, our goal is to learn $\mathcal{C}$ that maximises agent accuracy without any weight update:

$$\mathcal{C}^* = \arg \max_{\mathcal{C}} \text{Acc}(f_{\mathcal{C}}; \mathcal{D}_{\text{val}}), \quad (2)$$

where the maximisation is over $\mathcal{C}$ only and π_θ remains frozen ($\nabla_\theta = 0$).

3.2 Evolution Loop

Each evolution iteration over $\mathcal{D}_{\text{train}}$ has three phases (Figure 2).

Diagnose. The agent (Eq. 1) solves $\mathcal{D}_{\text{train}}$ with the current $\mathcal{C}$, then samples a sub-training set $\mathcal{D}_{\text{sub}} \subseteq \mathcal{D}_{\text{train}}$ containing both correct and incorrect attempts. The ANALYZERDECIDER inspects $\mathcal{D}_{\text{sub}}$ —the questions, original images $\mathbf{v}_i$, reasoning traces τ_i , and any intermediate tool outputs—and emits a root-cause analysis grounded in visual evidence, an identified bottleneck type (*cognitive* when the visual evidence is available but the reasoning procedure is weak, or *perceptual* when the agent needs a transformed visual input), and an action $a \in \{\text{skill}, \text{tool}, \text{both}\}$. Visual input here is essential: deciding whether a crop is misaligned,

whether labels are legible, or whether a processed image introduced artefacts requires inspecting the image, not just the reasoning trace.

Explore. Conditioned on the diagnosis, the GENERATOR proposes M candidate skill+tool combinations, each addressing the diagnosed bottleneck with a different strategy or implementation.

Validate and promote. Each candidate is evaluated on $\mathcal{D}_{\text{train}}$ and the highest-accuracy combination is promoted:

$$m^* = \arg \max_{m \in \{1, \dots, M\}} \text{Acc}(f_{\mathcal{C} \cup (s_m, t_m)}; \mathcal{D}_{\text{train}}),$$

$$\mathcal{C} \leftarrow \mathcal{C} \cup \{(s_{m^*}, t_{m^*})\}.$$
(3)

Eq. 3 is a training-set proxy for the validation objective in Eq. 2 and subsumes both an *origin* requirement (the new capability must improve over the previous $f_{\mathcal{C}}$ on the cases it targets) and a *regression* requirement (it must not degrade currently-correct cases). Because $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$ are disjoint, this proxy does not contaminate the held-out numbers we report; the only residual concern is selection-induced bias in m^* over a discrete candidate set, which scales logarithmically in M . We keep M at a single-digit candidate budget so this term stays small, and the $M=1$ ablation in Appendix B.4 measures the effect on held-out accuracy directly. When the action is both, the paired skill additionally serves as the tool’s mastery SOP, gating tool invocation at inference (Section 3.4). The full procedure runs for N iterations (default $N=3$); pseudocode is given in Appendix A (Algorithm 1).

3.3 Skills

A *skill* is a structured Markdown document with four fields: **When-to-Use** (a trigger predicate over question and image type), **Strategy** (a numbered step-by-step SOP), **Common Pitfalls**, and **Worked Example**. The Generator writes a skill conditioned on the AnalyzerDecider’s root-cause analysis; if a skill covering the same problem class already exists in $\mathcal{S}$, new insights are merged into it rather than creating a duplicate, preventing library bloat. At inference, the Solver retrieves the top- K skills by BM25 similarity to the question and appends them to the system prompt.

3.4 Tools and Mastery

A *tool* is a Python function (≤ 150 lines) that takes one or more image paths and returns processed images or extracted data, using standard CV libraries

(OpenCV, PIL, NumPy). The Generator writes a tool conditioned on a perceptual bottleneck diagnosis; representative examples produced in our experiments include chart re-rendering with enlarged axis labels, saliency-guided sub-region extraction, and contrast enhancement for low-quality documents.

Mastery as a paired skill. When the AnalyzerDecider’s action is both, the Generator emits a skill alongside the tool, and this paired skill is the tool’s mastery SOP. Its When-to-Use predicate specifies the input patterns on which the tool helps, its Common Pitfalls flag the patterns on which it hurts, and its Strategy instructs the Solver how to invoke the tool and consume its output. Because tool invocation is gated by retrieving the paired skill, deployment is selective by construction rather than indiscriminate—a property that distinguishes DYNAMO from prior tool-creation methods (Qian et al., 2023; Cai et al., 2024) that expose generated tools without learned applicability boundaries.

External tool sets (Mode B). When a curated tool set $\mathcal{T}_0 \neq \emptyset$ is already provided, DYNAMO can operate in **Mode B**: skip tool generation and instead synthesise a mastery skill for each provided tool $t_j \in \mathcal{T}_0$, learning when to invoke it from the agent’s own behaviour on $\mathcal{D}_{\text{train}}$. Mode B requires no code generation, making it practical when tool quality is already high but deployment strategy is unknown. Experiment II (Section 4.3) evaluates this mode.

3.5 Online Adaptation to Distribution Shift

In real deployments, DYNAMO runs against a streaming query distribution that can shift over time—a stream may begin with high-resolution perception cases and later start including MathVista-style reasoning cases, or vice versa. We design DYNAMO to detect such shifts and *update* its skills and tools online, without weight updates and without the practitioner having to anticipate the shift or prepare a per-family training set in advance.

Feedback signal. Running the evolution loop online requires a per-case correctness signal to drive diagnosis and selection (Eqs. 1–3). Raw production traffic typically lacks ground-truth labels, so DYNAMO assumes one of the practical feedback channels available in real deployments: a small fraction of queries reviewed by humans in the loop, an automated quality-assurance pipeline, or an LLM-based verifier that scores the agent’s answer post-hoc.

Our experiments use the benchmark’s ground-truth labels as a stand-in for this channel; the adaptation policy itself is agnostic to the source of the correctness signal, and the framework therefore does not claim fully unsupervised online adaptation.

DYNAMO continuously samples a sub-training set from the most recent queries (together with their correctness signals) and runs the evolution loop of Section 3.2 on them, accumulating skills and tools into the library $\mathcal{C}$. Let W be a rolling-window length. The dominant task family at step t is

$$\hat{f}_t = \arg \max_f \sum_{i=t-W+1}^t \mathbf{1}[\text{fam}(x_i) = f], \quad (4)$$

where $\text{fam}(\cdot)$ classifies each query from its question and image features. When Eq. 4 flips ($\hat{f}_t \neq \hat{f}_{t-1}$), the next evolution iteration is triggered on the new sub-stream and produces updated skills and tools that handle the new distribution. The library is thus kept in sync with the current query distribution. We treat this autonomous online adaptation—decoupled from the source of the correctness signal—as one of the method’s contributions; Experiment I (Section 4.2) evaluates it against a static baseline that never refreshes and an oracle whose full capability set is available from the start.

4 Experiments

4.1 Experimental Setup

Benchmarks. For the evolution-from-scratch setting (Exp I) we use four visual reasoning benchmarks: **ChartQA** (Masry et al., 2022) (multi-step numerical reasoning over charts; 2,500 test questions, relaxed accuracy with 5% tolerance), **Math-Vista** (Lu et al., 2024) (math reasoning grounded in figures and plots; 1,000 testmini questions, accuracy), **HRBench** (Wang et al., 2025) (perception on $\geq 4K$ -resolution images; accuracy), and **V*** (Wu and Xie, 2024) (visual search for an object’s property; 191 questions, accuracy); the first two stress *cognitive* bottlenecks while the latter two stress *perceptual* ones. For the tool-mastery setting (Exp II) we use **GTA** (Wang et al., 2024b), a benchmark of 229 real-world tasks with tools across perception, operation, logic, and creativity categories. GTA reports step-by-step accuracy on tool selection, argument prediction, and instruction following, plus an end-to-end answer accuracy. For the RL comparison (Exp III) we follow the VTool-R1 protocol (Wu

et al., 2025a) on its **Chart** and **Table** reasoning splits and reuse **V*** and **HRBench4K** as the high-resolution splits of the DeepEyes protocol (Zheng et al., 2025).

Foundation models. Experiment I evaluates five proprietary and open-weight VLMs: **GPT-4o**, **o4-mini**, **GPT-5.4**, **Doubao-Seed-2.0**, and **Qwen3.5-27B**. Experiment II evaluates **GPT-4o**, **GPT-5.4**, and **Doubao-Seed-2.0** on the GTA benchmark, plus a controlled **Doubao-Seed-2.0-Pro** analysis that inspects how provided-tool skills change with tool abstraction and task environment. Experiment III uses the **Qwen2.5-VL** family (3B, 7B) to align with the VTool-R1 evaluation protocol. All VLM weights are *frozen* throughout; no gradient updates are performed.

Evolution protocol. For each benchmark, we sample 20% of the official training split as $\mathcal{D}_{\text{train}}$ and run the DYNAMO evolution loop for $N=3$ iterations. The capability set $\mathcal{C}$ is then frozen and evaluated on the held-out validation / test split. For each evolution configuration (*Skill Only*, *Tool Only*, *Full*), we run **three independent seeds** that vary the 20% training subset and the Generator’s sampling, and report mean $\pm$ standard deviation in Table 1. The *None* (Base Agent) row invokes the frozen VLM with no evolved capabilities and is run once per cell—there is no capability-generation randomness to average over—so it carries no standard deviation.

Baselines. For Experiment I, we compare four evolution configurations that isolate the contribution of each component: **None (Base Agent)** invokes the frozen VLM with no evolved capabilities (the zero-shot baseline); **Skill Only** restricts the Generator to producing reasoning skills (no tool generation); **Tool Only** restricts the Generator to producing executable visual tools (no skill generation); **Full** is the unrestricted DYNAMO system, where the AnalyzerDecider chooses on each case whether to generate a skill, a tool, or both.

Evaluation metrics. All primary metrics are task-level accuracy on the held-out split. For Experiment II we report four of GTA’s five canonical metrics: three step-by-step metrics—**InstAcc** (Instruction-Following Accuracy: fraction of steps executed without errors), **ToolAcc** (Tool Selection Accuracy), and **ArgAcc** (Argument Prediction Accuracy)—and the end-to-end **AnsAcc** (Answer Accuracy on full tool-using execution). For the

supplementary controlled GTA analysis, we also report tool usage rate, average number of tool calls, and dominant tool-call chains. Experiment III follows the VTool-R1 evaluation protocol (Wu et al., 2025a) on Chart and Table reasoning splits, and uses DeepEyes (Zheng et al., 2025) as the RL reference for high-resolution V* and HRBench4K visual-search tasks.

4.2 Experiment I: Autonomous Evolution from Scratch

We test whether DYNAMO can bootstrap a useful capability library from scratch ($\mathcal{S}=\emptyset$, $\mathcal{T}=\emptyset$) across five backbones and four benchmarks, and whether the library remains useful when the query distribution shifts over time (Section 3.5).

Per-benchmark results. Table 1 reports accuracy for the four configurations on each backbone. Comparing the Full system to the zero-shot *None* baseline, DYNAMO improves direct inference on all 20 model–benchmark cells, with an average gain of +4.7 accuracy points. The largest gains are on o4-mini / HRBench (+12.7), GPT-5.4 / HRBench (+10.7), Doubao-Seed-2.0 / V* (+8.8), and Qwen3.5-72B / HRBench (+8.2). The relative strength of *Skill Only* and *Tool Only* also follows the cognitive / perceptual split: *Skill Only* is competitive on ChartQA and MathVista, where the gain comes from better decomposition and calculation procedures, while *Tool Only* and *Full* dominate on HRBench and V*, where sub-region inspection and visual search are central. On a few cells (GPT-4o / MathVista, Doubao / V*) *Skill Only* or *Tool Only* slightly edges out *Full*—an expected consequence of leaving capability choice to the AnalyzerDecider rather than enforcing both per case.

Capability quality. Appendix D inspects three representative forms of evolved capability: a ReFocus-style chart/table editing case, a ZoomEye-style coarse-to-fine search case, and an interleaved skill that pairs a textual SOP with visual evidence. The case studies illustrate what DYNAMO generates; they do not claim to reproduce the full algorithms of ReFocus (Fu et al., 2025) or ZoomEye (Shen et al., 2025).

Online adaptation to distribution shift. To simulate real online user traffic—where the task distribution grows and shifts over time as different users and requests arrive—we replay a 208-step non-stationary stream in which each phase intro-

duces a new task family while keeping carry-over cases from earlier phases: starting with HRBench, then adding MathVista, ChartQA, and finally V* as the new dominant family. The replay reuses completed per-case outputs, and the benchmark’s ground-truth labels stand in for the per-case correctness signal that a real deployment would obtain from human review, automated QA, or an LLM verifier (Section 3.5). This isolates the adaptation policy from the cost of regenerating capabilities online and from the cost of the feedback channel itself; we do not claim fully unsupervised online adaptation. We compare *Direct* (no capabilities), *Static* (capabilities evolved on the initial distribution only and never updated), *Online adapt* (DYNAMO detects the shift and updates skills and tools on the new sub-stream), and *Oracle* (capabilities pre-evolved on every family, an upper bound).

Figure 3 compares the four policies across all five backbones and three stream constructions: the *natural* mixture (a fixed-seed mix of the four families), *capability-relevant* (cases whose outcome depends on the matching capability), and *stress* (cases the matching capability is required to fix). The ordering $Direct \leq Static < Online\ adapt \approx Oracle$ is consistent across every backbone $\times$ protocol cell. On GPT-5.4 the gap is representative: *Online adapt* reaches 0.83 vs. 0.74 static on natural; widens to 0.89 vs. 0.65 on capability-relevant; and on stress, *Direct* collapses to 0.03 while *Online adapt* still recovers to 0.95. The per-step trajectory along the stream, with shift-detection latencies of 2–7 cases per phase, is shown in Appendix C.3 (Figure 5).

4.3 Experiment II: Learning to Use Pre-Provided Tools

Results on GTA. Table 2 compares three methods on the 121-case GTA validation split: *Base Agent* (zero-shot VLM with the GTA tools exposed); an *XSkill-style* baseline (Jiang et al., 2026) that injects a generic visual-reasoning skill header on top of the same tool environment but without GTA-specific protocols; and DYNAMO Mode B (skills + mastery learned on the training set). DYNAMO improves every metric on every backbone, with the largest gains on Doubao-Seed-2.0 (+18.7 ToolAcc, +15.8 ArgAcc, +16.1 InstAcc, +4.9 AnsAcc) and smaller but uniformly positive gains on GPT-4o (+4.3 / +2.2 / +11.4 / +5.0) and GPT-5.4 (+6.8 / +3.2 / +6.1 / +1.7).

Model	Evolution Strategy	HRBench	MathVista	V*	ChartQA
GPT-4o	None (Base Agent)	0.6386	0.6711	0.5828	0.7552
	Skill Only	0.6733±0.0121	0.7122±0.0038	0.6424±0.0066	0.7785±0.0026
	Tool Only	0.6452±0.0044	0.6900±0.0029	0.6225±0.0066	0.7722±0.0029
	Full	0.6886±0.0071	0.7189±0.0029	0.6689±0.0239	0.7799±0.0045
o4-mini	None (Base Agent)	0.73	0.753333	0.72847	0.7833
	Skill Only	0.7724±0.0058	0.7470±0.0107	0.7616±0.0066	0.7866±0.0151
	Tool Only	0.7457±0.1670	0.7356±0.0328	0.8387±0.0629	0.7920±0.0213
	Full	0.8571±0.0099	0.7559±0.0055	0.8698±0.0076	0.8007±0.0081
gpt5.4	None (Base Agent)	0.7471	0.7589	0.7748	0.7974
	Skill Only	0.7805±0.0036	0.7752±0.0039	0.7682±0.0199	0.8090±0.0022
	Tool Only	0.8367±0.0346	0.7663±0.0061	0.8477±0.0132	0.8108±0.0052
	Full	0.8538±0.0036	0.7815±0.0101	0.8521±0.0233	0.8276±0.0278
Doubao-Seed-2.0	None (Base Agent)	0.8214	0.8544	0.8675	0.8152
	Skill Only	0.8600±0.0049	0.8556±0.0109	0.8609±0.0115	0.8198±0.0031
	Tool Only	0.9005±0.0022	0.8659±0.0061	0.9448±0.0038	0.8156±0.0068
	Full	0.8990±0.0064	0.8681±0.0105	0.9558±0.0038	0.8347±0.0295
Qwen3.5-72B	None (Base Agent)	0.8171	0.8088	0.88079	0.8114
	Skill Only	0.8721±0.0030	0.8152±0.0080	0.8675±0.0066	0.8097±0.0090
	Tool Only	0.8986±0.0000	0.8241±0.0034	0.9514±0.0101	0.8115±0.0041
	Full	0.9010±0.0008	0.8252±0.0023	0.9603±0.0175	0.8158±0.0026

Table 1: **Autonomous evolution from scratch.** Accuracy across five VLM backbones and four benchmarks. The Full DYNAMO configuration improves over the zero-shot *None* baseline on all 20 model–benchmark cells.

What does environment-specific mastery contribute? The *XSkill-style* baseline isolates generic skill injection from environment-specific mastery: it uses the same tools and scorer but its skill is a generic visual-reasoning prompt without GTA’s tool-chain, argument, or answer-normalisation protocols. On every backbone, DYNAMO beats *XSkill-style* by 11–20 points on each step-level metric and 2.5–4.9 on AnsAcc. More tellingly, *XSkill-style* itself drops below *Base* on the step-level metrics for GPT-4o and GPT-5.4—adding a generic visual-reasoning prompt without learning the environment’s tool protocol can actually confuse tool selection. The gain over both baselines is therefore attributable not to skill injection per se, but to mastery skills that encode the target environment’s specific tool chains, argument formats, and answer normalisation rules.

Controlled tool-policy analysis. Table 3 runs a complementary controlled study on GTA with Doubao-Seed-2.0-Pro to inspect what the learned policy actually does when either the tool abstraction or the task environment changes. Two findings stand out. First, fixing the environment to OCR+Calculator, stronger composite tools compress the policy: atomic tools reach 86.7 via an OCR→Calculator chain (22/30 cases), while

Model	Method	ToolAcc	ArgAcc	InstAcc	AnsAcc
GPT-4o	Base Agent	80.3	62.0	52.0	66.1
	XSkill-style	71.3	53.0	49.5	68.6
	DYNAMO (Ours)	84.6	64.2	63.4	71.1
GPT-5.4	Base Agent	79.6	60.2	57.0	72.7
	XSkill-style	66.3	48.0	46.2	71.1
	DYNAMO (Ours)	86.4	63.4	63.1	74.4
Doubao-Seed-2.0	Base Agent	67.7	45.5	41.6	76.9
	XSkill-style	72.8	48.4	44.4	76.9
	DYNAMO (Ours)	86.4	61.3	57.7	81.8

Table 2: **Learning to use pre-provided tools on GTA (121-case validation split).** DYNAMO Mode B (skills + mastery only; no new tool generation) improves every metric on every backbone and beats both *Base Agent* and the *XSkill-style* (Jiang et al., 2026) generic-skill-injection baseline. *XSkill-style* itself drops below *Base* on the step-level metrics for GPT-4o and GPT-5.4, showing that generic skill injection without environment-specific protocols can confuse tool selection.

VisualArithmeticSolver reaches 90.0 with a one-call policy (30/30). Second, the same OCR tool plays different roles in different environments: in OCR+Calc it feeds arithmetic (OCR→Calculator, 19/20), while in OCR+Search it feeds external lookup (OCR→Search→OCR→Search, 20/20).

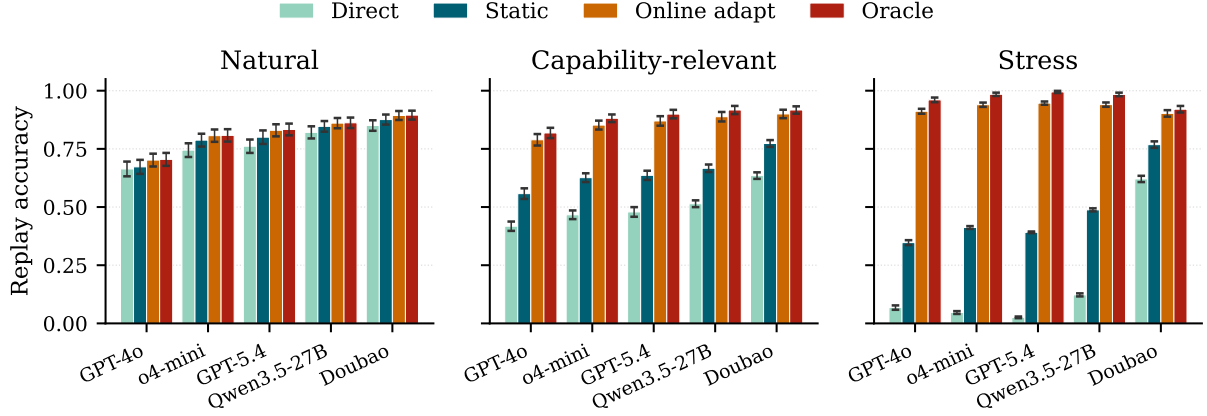


Figure 3: **Online adaptation to distribution shift.** Replay accuracy across five backbones and three stream constructions. *Online adapt* stays close to the *Oracle* upper bound and far above *Static* everywhere; the gap is largest under the stress protocol, where *Direct* nearly collapses. Error bars are seed std (50–200 seeds per cell).

Study	Condition	Cases	Acc.	Tool use	Avg. calls
Tool strength	Direct, no tools	30	83.3	0.0	0.00
Tool strength	Atomic OCR+Calc	30	86.7	100.0	1.83
Tool strength	Composite VAS	30	90.0	100.0	1.00
OCR role	OCR+Calc env	20	80.0	100.0	2.20
OCR role	OCR+Search env	20	85.0	100.0	2.35

Table 3: **Controlled tool-policy analysis on GTA (Doubao-Seed-2.0-Pro).** Top: stronger composite tools compress the policy at higher accuracy. Bottom: the same OCR tool exhibits different downstream behavior in different environments (dominant call chains discussed in prose). Acc./Tool use in %; Avg. calls is a count.

4.4 Experiment III: RL and Self-Evolution Across Visual Tool Tasks

We compare DYNAMO against task-specific RL on three task families using two Qwen2.5-VL backbones (3B, 7B): Chart and Table reasoning under the VTool-R1 protocol (Wu et al., 2025a), and high-resolution V* and HRBench4K under the DeepEyes protocol (Zheng et al., 2025).

Chart and Table reasoning. Using the gap-recovery fraction (DYNAMO – Direct)/(RL – Direct), DYNAMO recovers 65.7% and 99.4% of the Chart gap at 3B and 7B respectively, and 67.5% and 91.4% of the Table gap. DYNAMO + RL further beats RL alone by 1–3 points on every reported cell, indicating that the two interventions compose additively.

High-resolution visual search. On V* and HRBench4K, DYNAMO alone matches or beats task-specific RL at both scales: it gains +6.3 and +1.0 over RL on V* (3B / 7B) and +4.0 and +0.7 on HRBench4K. Adding DYNAMO on top of the

RL-trained policy raises 7B V* to 92.7 and 7B HRBench4K to 81.3, the strongest configurations in those cells.

Cost. The RL baselines require substantial training compute, and the protocol is the same across the two backbones in each family. VTool-R1’s Chart and Table checkpoints (both 3B and 7B) are each reported after 80 GRPO steps with a batch of 32 prompts and $n_{\text{rollout}}=8$ rollouts per prompt—approximately $80 \times 32 \times 8 = 20,480$ trajectories and 80 gradient updates per backbone. DeepEyes’ V* and HRBench4K checkpoints (both 3B and 7B) are each reported after 240 RL steps with a batch of 128 prompts and $n_{\text{rollout}}=4$ —approximately $240 \times 128 \times 4 = 122,880$ trajectories and 240 gradient updates per backbone. DYNAMO, by contrast, performs no weight updates: a run uses 20% of each benchmark’s training split for $N=3$ passes and invokes the frozen VLM only for solving, diagnosis, generation, and validation (a few thousand frozen-VLM API calls per benchmark; see Appendix A for the per-benchmark token totals). Across these three task families, DYNAMO therefore recovers most of the RL gain on Chart/Table and matches or beats RL on high-resolution visual search at a fraction of the training-time compute, while composing additively with RL when both are available.

4.5 Ablation Studies

The *Skill Only* and *Tool Only* rows of Table 1 already isolate the contribution of each capability type, so we use the remaining ablation budget to study the *mechanisms* that drive those gains rather

Backbone	Variant	Chart	Table	V*	HRBench4K
Qwen2.5-VL-3B	Direct	45.3995	43.0921	71.20	63.62
	RL	66.95	56.25	78.01	66.50
	DYNAMO	59.56	51.97	84.29	70.50
	DYNAMO + RL	68.28	59.54	85.34	71.25
Qwen2.5-VL-7B	Direct	56.0532	56.9078	79.06	70.50
	RL	75.18	68.42	84.82	76.88
	DYNAMO	75.060	67.43	85.86	75.12
	DYNAMO + RL	76.88	69.41	92.67	81.25

Table 4: **Self-evolution vs. task-specific RL across three task families.** Accuracy (%) on Qwen2.5-VL 3B and 7B. DYNAMO alone matches or beats RL on every high-resolution cell and recovers most of the Chart/Table gap; DYNAMO + RL is additive in every cell.

than rerunning the same capability-type comparison. We ablate six mechanisms inside the evolution loop: generic prompting vs. evolved capability, failure-only vs. both-correct-and-incorrect diagnosis, text-only vs. multimodal diagnosis, single-candidate vs. multi-candidate exploration, unpaired vs. paired mastery skill at tool invocation, and per-case vs. persistent capability storage. Appendix B reports the completed artifact-backed HRBench mechanism probe, plus a matched 200-case rerun audit for the remaining interventions.

5 Conclusion

DYNAMO is a training-free framework in which a frozen VLM agent builds and updates its own task-family-specific capability library from a small labeled training subset. The agent inspects both correct and incorrect attempts on a sampled sub-training set, an evolution planner decides whether to evolve a reasoning skill, an executable visual tool, or both, and the best of multiple candidates is promoted by validation accuracy. A mastery skill paired with each tool gates its invocation at inference, and an online adaptation loop updates the library when the query distribution shifts.

Across four visual reasoning benchmarks and five VLM backbones, DYNAMO improves direct inference on all 20 model–benchmark cells. On the tool-mastery benchmark GTA, the same framework lifts step-level tool selection, argument, and instruction-following accuracy across every tested backbone. Compared with task-specific RL (VTool-R1, DeepEyes) on chart, table, and high-resolution perception tasks, DYNAMO matches or beats RL on the high-resolution side and recovers most of the gap on the others, while composing additively when RL is also available. Together, these results

suggest that a substantial part of task-specific VLM adaptation can be obtained without any gradient updates, by letting the agent shape its own capability library.

Limitations

Short evolution horizon. We report results at $N=3$ iterations; the evolution dynamics suggest gains plateau by $N=5$, but we have not evaluated whether longer runs (e.g. $N>10$) eventually overfit the training distribution.

Tool quality ceiling. Tool effectiveness is bounded by the backbone’s Python coding ability. For benchmarks requiring specialised computer-vision operations beyond standard libraries (e.g. learned feature extractors), the Generator may fail to produce a working tool, limiting the perceptual gains.

Benchmark scope. We evaluate four visual-reasoning benchmarks plus GTA and the VTool-R1 / DeepEyes splits. Broader coverage (video QA, 3D spatial reasoning, medical imaging) is needed before strong generality claims about the cognitive / perceptual decision rule.

API cost. Each training case incurs roughly 15K tokens for the combined attempt, diagnosis, generation, and validation pass. While negligible compared with RL training, this cost should be accounted for when comparing training-free methods.

Reliance on a per-case correctness signal. Both the offline evolution loop and the online adaptation policy (Sections 3.2–3.5) require a per-case correctness signal to drive diagnosis and the multi-candidate selection rule. Our experiments use ground-truth labels from the benchmark splits as this signal, which corresponds to deployment settings with human-in-the-loop review, an automated quality-assurance pipeline, or an LLM-based post-hoc verifier. Deployments that have no access to any of these channels—i.e. fully unsupervised production traffic—fall outside the scope of the current method; extending DYNAMO to that setting would require a confidence-based or self-consistency-based feedback substitute that we leave to future work.

References

- Tianle Cai, Xuezhi Wang, Tengyu Ma, Xinyun Chen, and Denny Zhou. 2024. Large language models as tool makers. In *ICLR*. OpenReview.net.
- Xingyu Fu, Minqian Liu, Zhengyuan Yang, John Corring, Yijuan Lu, Jianwei Yang, Dan Roth, Dinei A. F. Florêncio, and Cha Zhang. 2025. Refocus: Visual editing as a chain of thought for structured image understanding. In *ICML*, Proceedings of Machine Learning Research. PMLR / OpenReview.net.
- Tanmay Gupta and Aniruddha Kembhavi. 2023. Visual programming: Compositional visual reasoning without training. In *CVPR*, pages 14953–14962. IEEE.
- Guanyu Jiang, Zhaochen Su, Xiaoye Qu, and Yi R. Fung. 2026. Xskill: Continual learning from experience and skills in multimodal agents. *CoRR*, abs/2603.12056.
- Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2024. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. In *ICLR*. OpenReview.net.
- Pan Lu, Baolin Peng, Hao Cheng, Michel Galley, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, and Jianfeng Gao. 2023. Chameleon: Plug-and-play compositional reasoning with large language models. In *NeurIPS*.
- Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq R. Joty, and Enamul Hoque. 2022. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *ACL (Findings)*, Findings of ACL, pages 2263–2279. Association for Computational Linguistics.
- Cheng Qian, Chi Han, Yi Ren Fung, Yujia Qin, Zhiyuan Liu, and Heng Ji. 2023. CREATOR: tool creation for disentangling abstract and concrete reasoning of large language models. In *EMNLP (Findings)*, Findings of ACL, pages 6922–6939. Association for Computational Linguistics.
- Runqi Qiao, Qiuna Tan, Minghan Yang, Guanting Dong, Peiqing Yang, Shiqiang Lang, Enhui Wan, Xiaowan Wang, Yida Xu, Lan Yang, Chong Sun, Chen Li, and Honggang Zhang. 2025. V-thinker: Interactive thinking with images. *CoRR*, abs/2511.04460.
- Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. Toolllm: Facilitating large language models to master 16000+ real-world apis. In *ICLR*. OpenReview.net.
- Haozhan Shen, Kangjia Zhao, Tiancheng Zhao, Ruochen Xu, Zilun Zhang, Mingwei Zhu, and Jianwei Yin. 2025. Zoomeye: Enhancing multimodal llms with human-like zooming capabilities through tree-based image exploration. In *EMNLP*, pages 6602–6618. Association for Computational Linguistics.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: language agents with verbal reinforcement learning. In *NeurIPS*.
- Alex Su, Haozhe Wang, Weiming Ren, Fangzhen Lin, and Wenhui Chen. 2025. Pixel reasoner: Incentivizing pixel-space reasoning with curiosity-driven reinforcement learning. *CoRR*, abs/2505.15966.
- Dídac Surís, Sachit Menon, and Carl Vondrick. 2023. Vipergpt: Visual inference via python execution for reasoning. In *ICCV*, pages 11854–11864. IEEE.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024a. Voyager: An open-ended embodied agent with large language models. *Trans. Mach. Learn. Res.*, 2024.
- Jize Wang, Zerun Ma, Yining Li, Songyang Zhang, Cailian Chen, Kai Chen, and Xinyi Le. 2024b. GTA: A benchmark for general tool agents. In *NeurIPS*.
- Wenbin Wang, Liang Ding, Minyan Zeng, Xiabin Zhou, Li Shen, Yong Luo, Wei Yu, and Dacheng Tao. 2025. Divide, conquer and combine: A training-free framework for high-resolution image perception in multimodal large language models. In *AAAI*, pages 7907–7915. AAAI Press.
- Mingyuan Wu, Jingcheng Yang, Jize Jiang, Meitang Li, Kaizhuo Yan, Hanchao Yu, Minjia Zhang, Chengxiang Zhai, and Klara Nahrstedt. 2025a. Vtoolrl: Vllms learn to think with images via reinforcement learning on multimodal tool use. *CoRR*, abs/2505.19255.
- Penghao Wu and Saining Xie. 2024. V*: Guided visual search as a core mechanism in multimodal llms. In *CVPR*, pages 13084–13094. IEEE.
- Rong Wu, Xiaoman Wang, Jianbiao Mei, Pinlong Cai, Daocheng Fu, Cheng Yang, Licheng Wen, Xueming Yang, Yufan Shen, Yuxin Wang, and Botian Shi. 2025b. Evolver: Self-evolving LLM agents through an experience-driven lifecycle. *CoRR*, abs/2510.16079.
- Xintong Zhang, Zhi Gao, Bofei Zhang, Pengxiang Li, Xiaowen Zhang, Yang Liu, Tao Yuan, Yuwei Wu, Yunde Jia, Song-Chun Zhu, and 1 others. 2025. Adaptive chain-of-focus reasoning via dynamic visual search and zooming for efficient vlms. *arXiv preprint arXiv:2505.15436*.
- Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. Expel: LLM agents are experiential learners. In *AAAI*, pages 19632–19642. AAAI Press.

Ziwei Zheng, Michael Yang, Jack Hong, Chenxiao Zhao, Guohai Xu, Le Yang, Chao Shen, and Xing Yu. 2025. Deepeyes: Incentivizing "thinking with images" via reinforcement learning. *CoRR*, abs/2505.14362.

A Implementation Details

Algorithm. Algorithm 1 gives the full DYNAMO evolution procedure described in Section 3.2.

Algorithm 1: DYNAMO evolution loop.	
Input: $\mathcal{D}_{\text{train}}$, frozen VLM π_θ , iterations N , candidates M	
Output: Capability set $\mathcal{C} = (\mathcal{S}, \mathcal{T})$	
1	$\mathcal{C} \leftarrow (\emptyset, \emptyset);$
2	for $n = 1$ to N do
3	Solve $\mathcal{D}_{\text{train}}$ with $f_{\mathcal{C}}$; collect attempts;
4	$\mathcal{D}_{\text{sub}} \leftarrow$ sample from $\mathcal{D}_{\text{train}}$;
5	$a, r \leftarrow$
	ANALYZERDECIDER($\mathcal{D}_{\text{sub}}$; π_θ);
6	$\{(s_m, t_m)\}_{m=1}^M \leftarrow$
	GENERATOR(a, r ; π_θ); // s_m
	paired with t_m acts as its
	mastery SOP
7	$m^* \leftarrow$
	$\arg \max_m \text{Acc}(f_{\mathcal{C} \cup \{(s_m, t_m)\}}; \mathcal{D}_{\text{train}});$
8	if accuracy improves then
9	$\mathcal{C} \leftarrow \mathcal{C} \cup \{(s_{m^*}, t_{m^*})\};$
10	end
11	end
12	return $\mathcal{C}$;

Role prompts. DYNAMO uses three role-specific system prompts for the same base VLM: *Solver* (standard QA prompt with capability context injected), *AnalyzerDecider* (root-cause analysis and decision), and *Generator* (skill or tool generation). All prompts are provided in full at [\[anonymised repository\]](#).

Skill retrieval. At inference time, the Solver retrieves the top- $K=3$ skills from $\mathcal{S}$ using BM25 cosine similarity over skill titles and When-to-Use fields. Retrieved skill text is appended to the system prompt.

Tool invocation. The Solver inspects each tool’s mastery SOP to decide whether to invoke it on the current input. If the input matches a tool’s supported patterns, the tool is applied and its output image(s) are appended to the multimodal context. Tool execution is sandboxed with a 30-second timeout and resource limits.

Hyperparameters. Table 5 lists all hyperparameter settings.

Table 5: Hyperparameter settings.

Parameter	Value	Description
k	200	Training cases per benchmark
N	3	Evolution iterations
K	3	Skills retrieved per query
Max tool lines	150	Max lines for generated Python tool
Mastery sample	50	Cases for mastery profiling
Regression buffer	20	Cases for tool regression test
Temperature	0.7	Generator temperature
Max tokens (gen.)	2048	Max tokens for capability generation

Compute. Experiments use the frozen VLM backbones listed in Section 4.1. Total API calls per benchmark: $\leq k \times N \times 3 = 1,800$ calls (solver + analyzer + generator, upper bound). Estimated token usage: $\sim 3\text{M}$ tokens per benchmark. No GPU training is performed.

B Mechanism Ablations

Experiment I in the main paper already isolates the contribution of each capability *type*: the *None*, *Skill Only*, *Tool Only*, and *Full* rows of Table 1 answer whether skills, tools, or both account for the observed gains. The ablations here test a complementary question: when the Full system is used, which *mechanisms* inside the evolution loop (Section 3.2) are doing the work? Each variant removes one design decision and holds the rest fixed, so each row of Table 6 rules out one alternative explanation for the Full system’s gains.

Setup. The full grid of mechanism ablations is expensive because each row rebuilds $\mathcal{C}$ from scratch. We therefore report the completed artifact-backed probes first, and use them to answer the most immediate reviewer question: is the gain merely generic prompting, or does the evolved visual-tool path matter? All rows in Table 6 use the same HRBench validation set (700 cases) and the same function-calling VQA runtime with Doubao-Seed-2.0-Pro. For the first three rows, tools are disabled, so the comparison isolates prompt/skill content; the last two rows reuse the existing full-val tool-enabled runs for context.

Completed mechanism probe. Table 6 shows that generic skill injection does not explain the HRBench gain. A neutral task prompt is slightly below the no-capability runtime, and the evolved no-tool skill is essentially tied with it. In contrast,

Variant	Mechanism isolated	Tools	Correct / Total	Accuracy
No Capability	no skill context; no tool exposure	off	599 / 700	0.8557
Generic Skill	evolved library → hand-neutral HRBench prompt	off	594 / 700	0.8486
Evolved No-Tool Skill	evolved HRBench SOP, but visual tools disabled	off	597 / 700	0.8529
Evolved Tool Only	validated visual tools/artifacts, no extra generated skill text	on	629 / 700	0.8986
Evolved Skill + Tool	paired evolved skill/tool deployment	on	625 / 700	0.8929

Table 6: **Completed HRBench mechanism probe.** The first three rows disable tools and test whether generic prompt/skill text explains the gain. It does not: generic and evolved no-tool skills are near the no-capability runtime. The tool-enabled rows show the much larger gain from validated visual artifacts, isolating the perceptual mechanism used by DYNAMO on HRBench.

enabling the evolved visual-tool/artifact path yields a much larger improvement. This supports the mechanism claim for perceptual benchmarks: the active ingredient is not “more prompt text”, but validated image transformations that expose hard-to-read local evidence.

The source artifacts for the first three rows are `ablate_hrbench_nocap_fc_doubao_20260520T074011Z`, `ablate_hrbench_generic_skillonly_fc_doubao_20260520T074011Z`, and `ablate_hrbench_evolved_skillonly_fc_doubao_20260520T074011Z`. The tool-enabled context rows are `exp1r1_doubao_hrbench_toolonly_20260519T140901Z_doubao_sourceplus2_w64` and `exp1r1_doubao_hrbench_skilltool_20260519T140901Z_doubao_sourceplus2_w64`.

Additional mechanism-rerun audit. We also reran the remaining mechanism interventions on HRBench with the same Doubao-Seed-2.0-Pro runtime, a 25-case evolution subset, and the first 200 validation cases under the structured benchmark ordering. This audit is useful diagnostically, but we do not use it as the headline evidence because the small training subset was already near-saturated: the no-capability adaptive solver solved 23/25 cases, leaving little promotion signal. In this setting, the Full, Failure-Only Diagnosis, Text-only Diagnosis, and No-Persistence variants proposed candidates but promoted no reusable capability. Single-Candidate and No Paired Mastery each promoted one round-3 skill-only fallback ($\Delta = +0.04$ on the 25-case subset). The No Paired Mastery run therefore did *not* instantiate the intended generated-tool-without-mastery test; its accepted bundle had zero generated tools.

Guarded 30-case tool-generation pilot. After adding the generated-tool acceptance guards, we

ran a time-bounded HRBench pilot on the first 30 validation cases to check whether the mechanism interventions now instantiate the intended tool path. This pilot is not a replacement for the 700-case mechanism probe above; it is an audit of the evolution loop under the stricter acceptance criterion. We therefore report the selection score used by the loop—candidate accuracy on the same 30-case subset against the same-run baseline—rather than treating the noisy final replay as a held-out estimate.

B.1 Generic Skill vs. Evolved Capability

What is removed. The evolved capability library is replaced with a single hand-neutral task prompt that gives generic task advice (for ChartQA: “read the chart carefully, identify relevant values, reason step by step”; for HRBench: “inspect local details before answering and verify the target object”). No diagnosis, generation, validation, or persistence is run.

What this rules out. A pure prompt-engineering account would predict that Generic Skill closes most of the gap to Full, since the evolved skills are themselves natural-language SOPs. If Generic Skill improves over Direct by only a small margin and stays well below Full, the gap is attributable to mechanism-level evolution—failure-specific procedures, applicability triggers, and pitfalls that a generic prompt cannot anticipate—rather than to the presence of any task-aware prompt.

Observed pattern. On HRBench with tools disabled, Generic Skill obtains 0.8486 accuracy, compared with 0.8557 for the no-capability runtime and 0.8529 for the evolved no-tool skill. These rows show that simply adding a neutral task prompt does not recover the tool-enabled gains. The tool-

Variant	Outcome on the matched 200-case HRBench subset	Correct / Total	Accuracy
No Capability	matched control from the 700-case artifact	172 / 200	0.8600
Generic Skill	matched generic-prompt control	171 / 200	0.8550
Evolved No-Tool Skill	matched no-tool evolved-skill control	170 / 200	0.8500
Evolved Tool Only	matched tool-enabled control	183 / 200	0.9150
Evolved Skill + Tool	matched paired deployment control	180 / 200	0.9000
Single-Candidate ($M=1$)	promoted one skill-only fallback in round 3	170 / 200	0.8500
No Paired Mastery	promoted one skill-only fallback; no generated tool was accepted	174 / 200	0.8700
Full rerun	no candidate promoted on the 25-case evolution subset	—	—
Failure-Only Diagnosis	no candidate promoted on the 25-case evolution subset	—	—
Text-only Diagnosis	no candidate promoted on the 25-case evolution subset	—	—
No Persistence	no candidate promoted on the 25-case evolution subset	—	—

Table 7: **HRBench mechanism-rerun audit on a matched 200-case subset.** The matched controls come from the completed 700-case artifacts but are filtered to the same case IDs as the 2026-05-22 mechanism rerun. The audit confirms that the two promoted fallback skills do not recover the tool-enabled gain, while the no-promotion rows should be read as negative diagnostics rather than held-out mechanism effect sizes.

enabled rows in Table 6 are roughly four points higher, so the HRBench improvement is better explained by validated visual artefacts than by generic skill injection.

B.2 Failure-Only Diagnosis

What is removed. The AnalyzerDecider only sees the incorrect cases in the sampled sub-training set; the correct cases are dropped. Everything else—multimodal input, multi-candidate exploration, paired mastery skill, persistence—is unchanged.

What this rules out. Most prior self-improvement work treats correct cases as nothing to learn from. This variant tests that assumption directly. If Failure-Only Diagnosis matches Full, then the correct cases carry no useful diagnostic signal and the abstract’s framing of “diagnoses both correct and incorrect attempts” is decorative. A gap supports the design choice: correct attempts tell the AnalyzerDecider *which sub-patterns the current C already handles*, which sharpens the next capability proposal toward what is still missing.

Observed pattern. In the guarded 30-case HRBench pilot, Failure-Only Diagnosis promoted a generated cross-position tool and paired mastery skill, improving the selection score from 0.6000 to 0.7000. This does not support a strong claim that

correct cases are always necessary; on this narrow perceptual cluster, the failures alone were sufficient to recover a useful tool. We therefore treat this row as evidence that failure-only diagnosis can work on concentrated HRBench cross-position errors, while leaving the broader benefit of correct-case contrast to larger runs.

B.3 Text-only Diagnosis

What is removed. The AnalyzerDecider’s inputs are restricted to text only: the question, the agent’s answer (correct or incorrect), the ground-truth answer, and the reasoning trace. No original image, no intermediate tool output, and no processed visual artefact is passed in.

What this rules out. Text reflection can say *that* an answer was wrong, but it cannot tell whether a crop was misaligned, whether a zoomed region missed the target, or whether a processed image introduced artefacts. We expect the gap to Full to be largest on the perceptual benchmarks (HRBench, V^*). On ChartQA the gap should be smaller because many failures there are procedural; a small ChartQA gap with a large HRBench/ V^* gap is itself the signature of the visual-diagnosis claim.

Observed pattern. The guarded HRBench pilot produced the expected failure mode: the text-only candidate was rejected because the proposed tool did not produce a visual artifact. Its selection score

Variant	Guarded outcome	Baseline	Candidate	Δ
Full tool-guard pilot	accepted one generated tool with its paired mastery skill; target HRBench-cross family improved from 0.5000 to 0.8125	0.5667	0.6000	+0.0333
Failure-Only Diagnosis	accepted one generated tool with paired mastery skill; failed cases alone exposed the cross-position bottleneck in this subset	0.6000	0.7000	+0.1000
Text-only Diagnosis	rejected: the proposed tool produced no visual artifact, so no generated tool was promoted	0.6667	0.6667	0.0000
Single-Candidate ($M=1$)	accepted one generated tool with paired mastery skill from a single proposal	0.5333	0.6000	+0.0667
No Paired Mastery	smoke passed, but the unpaired tool candidate regressed on the full subset; HRBench-cross fell from 0.5625 to 0.4375	0.5667	0.5333	-0.0333
No Persistence	the candidate regressed before persistence mattered, so this row is diagnostic but not a clean persistence ablation	0.5667	0.5333	-0.0333

Table 8: **Guarded HRBench tool-generation pilot on 30 validation cases.** All rows require at least one generated tool and forbid skill-only fallback. The Full row uses the accepted tool-guard pilot ablation_hrbench_toolguard_fastdry30_w8_v5_doubao_20260523T110300Z; the other rows use the matched one-round guarded variants ablation_hrbench*_fastdry30_w3_doubao_20260523T114000Z. These numbers are mechanism-audit scores, not held-out effect sizes. A same-w3 Full replicate was started but omitted because Doubao returned sustained TPM-limit errors during candidate evaluation after smoke validation.

therefore stayed at the same-run baseline (0.6667 to 0.6667) and no generated tool was promoted. This supports the qualitative mechanism claim that visual diagnosis is needed to materialize useful image-processing tools, not just to write another textual rule.

B.4 Single-Candidate ($M=1$)

What is removed. The Generator produces a single capability proposal per iteration (the same setting as typical generate-then-validate tool-creation methods such as CREATOR (Qian et al., 2023) and LATM (Cai et al., 2024)). The validate-and-promote step (Eq. 3) collapses to a binary accept/reject of that single candidate against the training set.

What this rules out. A reviewer might ask whether the multi-candidate $\arg \max$ in Eq. 3 is load-bearing or whether the first proposal is usually good enough. If Single-Candidate roughly matches Full, the exploration step is decorative. A measurable gap shows that diverse candidates plus selection by training-set accuracy is what turns a noisy proposal distribution into a reliably improving library.

Observed pattern. On the guarded 30-case HRBench pilot, the single-candidate run accepted one generated lateral-position tool and paired mastery skill, improving from 0.5333 to 0.6000. This

is a positive result for the generator, but it is not evidence that multi-candidate selection is unnecessary: the accepted capability targets the same narrow HRBench-cross bottleneck where several runs generated similar position-marking tools. We therefore report this row as a successful small-subset stress test of the guarded tool path, not as a settled multi-candidate ablation.

B.5 No Paired Mastery Skill

What is removed. When the AnalyzerDecider’s action is both, the Generator emits only the tool and not its paired mastery skill. The tool is added to $\mathcal{T}$ but has no WHEN-TO-USE predicate gating its invocation at inference, so any skill retrieval that surfaces a tool reference exposes the tool unconditionally.

What this rules out. Prior tool-creation methods deploy generated tools indiscriminately. The paired-skill design in Section 3.4 predicts that gating tool invocation by retrieval of an applicability-encoding skill is what keeps tool usage selective. No Paired Mastery Skill is expected to keep or even increase tool-usage rate but reduce accuracy on benchmarks where naive tool exposure can hurt—MathVista is the canonical cautionary case.

Observed pattern. The guarded HRBench pilot cleanly instantiated this ablation. The unpaired generated tool passed smoke validation, but once

exposed without a mastery skill it regressed on the 30-case subset (0.5667 to 0.5333) and was rejected. The targeted HRBench-cross family also dropped from 0.5625 to 0.4375. This is the strongest new mechanism evidence from the pilot: the tool itself can be executable and locally plausible, while the missing when-to-use gate still makes it harmful at subset scale.

B.6 No Persistence

What is removed. Capabilities promoted during one iteration are not stored in $\mathcal{C}$. The agent can still reflect on a case and immediately retry it with the generated capability, but the capability is discarded before the next case is processed, so no cross-case retrieval occurs.

What this rules out. A retry-only account would predict that most of Full’s gain comes from in-place retry on the triggering case rather than from cross-case reuse. If No Persistence matches Full on held-out accuracy, the library is a bookkeeping device with no inferential value. A gap shows that the system’s advantage comes from accumulating capabilities that future cases can retrieve, not just from one more attempt at the triggering case. ChartQA is the cleanest test because failures there form recurring families (stacked-bar boundary reading, multi-series selection) that a single capability can fix across many held-out cases.

Observed pattern. The guarded HRBench pilot did not produce a clean persistence test. The generated candidate regressed before the persistence switch mattered (0.5667 to 0.5333), so the run was rejected under the same rule as ordinary candidate selection. We therefore mark this row as inconclusive rather than claiming evidence for or against cross-case persistence. The persistence question still needs a row where the same generated capability improves the triggering subset and is then explicitly discarded before later cases.

B.7 Sensitivity to Training Subset Size and Iteration Count

The main paper fixes the training subset at 20% of each benchmark’s training split and runs $N=3$ evolution iterations. A sensitivity sweep over smaller fractions (e.g. 5%, 10%) and over $N \in \{1, 3, 5\}$ tests how quickly the library saturates: a small subset should already expose the most frequent procedural / perceptual patterns, with diminishing

returns as the subset grows; $N=1$ should miss capabilities that only surface after earlier ones reshape the trajectory, while $N=3$ and $N=5$ should produce comparable libraries. We treat this as a robustness check rather than a tuning study; the operating point is chosen to fit the per-benchmark budget.

B.8 Summary

Read together, the mechanism ablations are designed to triangulate the same claim from different angles. The completed HRBench probe already rules out a pure prompt-engineering explanation and points to the visual-tool/artifact path as the active mechanism on perceptual tasks. The additional 200-case rerun audit is consistent with that conclusion: the promoted Single-Candidate and No-Paired-Mastery fallbacks behave like no-tool skills and stay below the matched tool-enabled controls, while several mechanism variants promoted no capability on a near-saturated 25-case training subset. We therefore treat Failure-Only Diagnosis, Text-only Diagnosis, No Paired Mastery Skill, and No Persistence as precisely defined intervention protocols, with the completed HRBench artifacts indicating which variants produced citeable held-out evidence under the current budget. The guarded 30-case pilot in Table 8 updates that audit with stricter acceptance checks: it confirms that the evolution loop can promote real generated tools, that text-only diagnosis fails to materialize a usable artifact, and that removing paired mastery can turn an executable tool into a harmful subset-level intervention. At the same time, it keeps the Failure-Only and No-Persistence conclusions modest, because those rows do not yet isolate the intended mechanisms under the current budget.

Practical diagnostic. The mechanism ablations also suggest a practical rule of thumb for new task families. If the bottleneck is procedural—visual evidence is present but the reasoning is weak—skill evolution is the lower-cost first choice; if a different view of the image is required before the evidence becomes accessible, tool evolution is necessary. A practitioner targeting a new benchmark can use a small pilot run (e.g. 5% of the training split with $N=1$) to inspect which kinds of capabilities are generated and whether they transfer beyond the triggering examples.

C Additional Results

C.1 Training Set Size Sensitivity



Figure 4: Placeholder for ChartQA accuracy vs. training set size $k \in \{10, 25, 50, 100, 200\}$ at $N=3$ iterations.

Figure 4 shows ChartQA accuracy as a function of training set size $k \in \{10, 25, 50, 100, 200\}$ at $N=3$ iterations.

C.2 Per-Category Analysis on ChartQA

C.3 Distribution-Shift Adaptation: Time Series and Aggregates

Figure 3 in the main paper aggregates the distribution-shift experiment across backbones and stream constructions. This appendix shows the per-step trajectory along the stream (Figure 5) and the full per-backbone numerical aggregates (Table 9).

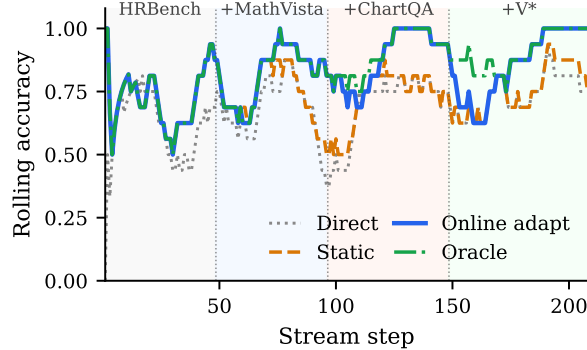


Figure 5: **Rolling accuracy along the natural stream (GPT-5.4).** Phase labels above the plot mark the newly introduced family at each shift; earlier families remain present in the mix. *Static* lags at each shift; *Online adapt* catches up to *Oracle* within a 2–7 case detection latency.

D Qualitative Analysis of Evolved Capabilities

This appendix gives three qualitative case studies of evolved capabilities. The examples are intended to make the generated artifacts concrete, not to introduce new quantitative results. Each case should be instantiated with the corresponding logged failure, generated skill/tool file, visual artifact, and

Replay protocol	Direct	Static	Online adapt	Oracle
<i>GPT-4o (Full)</i>				
Capability-relevant	0.418±0.020	0.558±0.023	0.789±0.025	0.818±0.022
Stress	0.068±0.009	0.347±0.011	0.912±0.008	0.961±0.010
Natural	0.664±0.032	0.673±0.030	0.702±0.028	0.705±0.028
<i>o4mini (Full)</i>				
Capability-relevant	0.467±0.019	0.626±0.019	0.852±0.019	0.881±0.016
Stress	0.047±0.006	0.412±0.006	0.940±0.009	0.984±0.007
Natural	0.745±0.029	0.787±0.028	0.807±0.027	0.808±0.027
<i>GPT-5.4 (Full)</i>				
Capability-relevant	0.479±0.021	0.636±0.020	0.870±0.020	0.899±0.018
Stress	0.026±0.004	0.391±0.004	0.947±0.007	0.995±0.005
Natural	0.761±0.029	0.800±0.029	0.830±0.026	0.833±0.025
<i>Qwen3.5-72B (Full)</i>				
Capability-relevant	0.514±0.015	0.667±0.016	0.888±0.020	0.917±0.018
Stress	0.122±0.007	0.488±0.007	0.940±0.009	0.984±0.008
Natural	0.821±0.026	0.847±0.023	0.861±0.022	0.862±0.022
<i>Doubao-Seed-2.0 (Full)</i>				
Capability-relevant	0.635±0.014	0.773±0.015	0.900±0.018	0.917±0.016
Stress	0.621±0.013	0.768±0.014	0.902±0.014	0.920±0.014
Natural	0.850±0.023	0.876±0.022	0.893±0.019	0.895±0.019

Table 9: **Per-backbone aggregates for the distribution-shift experiment.** Mean accuracy with seed std (50–200 seeds per cell). Bars in Figure 3 are computed from this table.

validation record before submission. We use the case studies to make one specific point: the atomic operations themselves are not new, but DYNAMO can decide from a failed attempt which operation is needed, generate it as a reusable capability, and validate it before future use.

D.1 Case A: ReFocus-Style Skill and Tool for Structured Images

ReFocus (Fu et al., 2025) proposes visual editing as a form of chain-of-thought for structured image understanding, by asking an MLLM to generate Python code that edits the input image—drawing boxes, highlighting relevant regions, masking distractors—before reasoning. The case below illustrates that DYNAMO can produce a capability with the same flavour *autonomously*: on a ChartQA training case, the evolution loop diagnosed a visual-disambiguation bottleneck and proposed the paired skill and tool shown next.

Generated skill (paired with the tool below; triggered on ChartQA):

```
## Skill: Edit-Then-Read for Structured Images

When-to-Use:
Use when the question refers to a specific chart/table element
that is visually close to distractors, such as one series
among
many lines, one segment in a stacked bar, or one table
row/column.

Strategy:
1. Identify the visual entity named by the question.
2. Before computing the answer, create an edited view that
   marks
   only the relevant region and suppresses likely distractors.
3. Re-read the edited image and extract the required value.
4. Perform the requested arithmetic or comparison.
```

Case study	Generated capability	Prior motif	Claim supported
Case A: Structured-image editing	A skill that instructs the solver to isolate the relevant chart/table structure, plus a tool that highlights, boxes, or masks visual evidence.	ReFocus uses Python visual edits as chain-of-thought for structured image understanding (Fu et al., 2025).	DYNAMO can generate a reusable visual-editing capability from failure, rather than relying on a fixed visual-editing interface.
Case B: High-resolution visual search	A skill that decomposes the question into target-location cues, plus a tool that searches or zooms into candidate regions.	ZoomEye performs training-free tree-based image exploration over zoomed sub-regions (Shen et al., 2025).	DYNAMO can generate a local coarse-to-fine search capability without being given ZoomEye as a predefined module.
Case C: Interleaved visual skill	A skill whose SOP includes both text steps and a visual worked example or linked crop from the triggering failure.	Visual-thought methods show that intermediate visual artifacts can guide structured reasoning.	DYNAMO can store procedural and visual evidence together as a retrieved capability.

Table 10: **Three qualitative forms of evolved capability.** The case studies compare generated artifacts to prior visual-reasoning motifs. They do not claim that DYNAMO reproduces the full prior algorithms.

Common Pitfalls:

- Do not answer from the unedited image if multiple marks are visually similar.
- Do not treat the visual edit as adding new information; it only exposes the evidence already present in the image.

Generated tool (paired with the skill above):

```
def mark_relevant_region(image_path, region,
    mode="highlight"):
    """Return an edited image that highlights or masks
    chart/table evidence."""
    from PIL import Image, ImageDraw
    img = Image.open(image_path).convert("RGB")
    draw = ImageDraw.Draw(img, "RGBA")
    x1, y1, x2, y2 = region
    if mode == "highlight":
        draw.rectangle([x1, y1, x2, y2], outline=(255, 0, 0,
            255), width=6)
        draw.rectangle([x1, y1, x2, y2], fill=(255, 255, 0,
            60))
    elif mode == "mask_distractors":
        mask = Image.new("RGBA", img.size, (255, 255, 255,
            180))
        ImageDraw.Draw(mask).rectangle([x1, y1, x2, y2],
            fill=(0, 0, 0, 0))
        img = Image.alpha_composite(img.convert("RGBA"), mask)
        out_path = image_path.replace(".png", "_marked.png")
        img.convert("RGB").save(out_path)
        return out_path
```

Like ReFocus, this capability uses visual editing as an intermediate reasoning step: the solver does not merely produce a textual explanation but creates an edited visual artefact that changes the evidence shown to the next solver call. The difference is that ReFocus defines visual editing as the reasoning interface up front, whereas DYNAMO emits this skill/tool pair only after the evolution loop’s diagnosis identifies visual disambiguation as the bottleneck on the triggering ChartQA case.

D.2 Case B: ZoomEye-Style Skill and Tool for High-Resolution Search

ZoomEye (Shen et al., 2025) addresses the loss of fine visual detail by treating image understanding as a tree-based exploration process. The full image is the root node, zoomed image regions are child nodes, and the model searches over promising regions until it finds the visual evidence needed to answer. A DYNAMO case belongs in this category when the generated capability performs region selection, cropping, zooming, or coarse-to-fine search.

Triggering failure. The triggering case should involve a high-resolution image where the base agent knows the object or region type but cannot inspect it accurately from the full image. Examples include small text, distant signs, number plates, small object attributes, or a target object occupying a small fraction of the image. [Insert benchmark, case id, question, ground-truth answer, and base answer.]

Generated skill. The evolved skill should make the search procedure explicit:

Skill: Coarse-to-Fine Region Search

When-to-Use:

Use when the question asks about a small object, distant text, fine-grained attribute, or localized region in a high-resolution image.

Strategy:

1. Parse the question for target object, attribute, and spatial cues.
2. Inspect the full image only to estimate candidate regions.
3. Divide the image into coarse tiles and score each tile for relevance.
4. Zoom into the best candidate region and answer from the enlarged view.
5. If the crop is ambiguous, repeat with a finer crop around the target.

Common Pitfalls:

- Do not answer from the globally downsampled image when the requested evidence is small.
- Do not crop solely by image center; use question-specific cues.

Generated tool. The paired tool implements a local search/zoom operation. The final version should quote the actual generated code; a representative form is:

```
def coarse_to_fine_zoom(image_path, target_description,
    grid_size=4):
    """Return candidate high-resolution crops for a
    target described in text."""
    from PIL import Image
    img = Image.open(image_path).convert("RGB")
    w, h = img.size
    crops = []
    for gy in range(grid_size):
        for gx in range(grid_size):
            box = (gx*w//grid_size, gy*h//grid_size,
                (gx+1)*w//grid_size, (gy+1)*h//grid_size)
            crop = img.crop(box)
            out = image_path.replace(".png",
                f"_tile_{gx}_{gy}.png")
            crop.save(out)
            crops.append((out, box))
    return crops
```

Why this matches ZoomEye. The similarity is the coarse-to-fine visual search motif. Both methods avoid forcing the VLM to answer from a single fixed-resolution full image. The difference is scope: ZoomEye is a general tree-search algorithm, while the DYNAMO capability is generated from a particular failure and may implement only the subset of zoom/search behavior needed for that task family.

Evidence to insert. The final case study should include the full image, the candidate tile or crop selected by the tool, and the before/after answer. **[Insert selected crop, validation result, and retry answer.]**

D.3 Case C: Interleaved Visual Skill

The third case is different from the previous two because the main artifact is not only an executable program. An interleaved visual skill is a retrieved Markdown skill that contains both textual reasoning instructions and visual evidence, such as an embedded crop, annotated chart, or linked processed image from the triggering failure.

Triggering failure. This case should involve a problem where a text-only SOP is underspecified because the visual pattern itself matters. A common example is chart reasoning: the agent may know that it should subtract segment boundaries in a stacked bar chart, but still confuse which boundary belongs to the target segment. **[Insert benchmark, case id, question, ground-truth answer, and base answer.]**

Generated skill. The generated skill should be quoted directly if available. A representative

interleaved form is:

```
## Skill: Stacked-Bar Segment Reading with Visual Anchor

When-to-Use:
Use when the question asks for a value, difference, or
comparison
involving a segment of a stacked bar chart.

Strategy:
1. Locate the target bar using the x-axis label.
2. Locate the target segment using the legend colour.
3. Read the lower and upper boundaries of the segment.
4. Subtract lower from upper to obtain the segment value.
5. Verify the result against the visual anchor below.

Visual Anchor:
[embedded crop or linked image showing the annotated segment
boundaries]

Common Pitfalls:
- Do not read the total bar height when the question asks for
one segment.
- Do not use a nearby segment with a visually similar colour.
```

Optional generated tool. In this case, the paired tool may be simple: it produces the crop or annotation that is embedded in the skill.

```
def make_visual_anchor(image_path, crop_box, annotations):
    """Create an annotated crop used as the visual anchor in
    a skill."""
    from PIL import Image, ImageDraw
    img = Image.open(image_path).convert("RGB")
    crop = img.crop(crop_box)
    draw = ImageDraw.Draw(crop)
    for ann in annotations:
        draw.rectangle(ann["box"], outline=ann.get("color",
            "red"), width=4)
        if "label" in ann:
            draw.text((ann["box"][0], ann["box"][1]),
                ann["label"],
                fill=ann.get("color", "red"))
    out_path = image_path.replace(".png",
        "_visual_anchor.png")
    crop.save(out_path)
    return out_path
```

Why this is interleaved. The retrieved capability is not purely textual and not purely executable. The skill tells the agent how to reason, while the visual anchor shows what the relevant visual configuration looks like. This is related to visual-thought methods, but the visual artifact is stored persistently inside the capability library and retrieved on later cases.

Evidence to insert. The final case study should include the generated Markdown skill, the visual anchor image, and a before/after answer showing how the retrieved interleaved skill changes the solver's behavior. **[Insert validation result and retry answer.]**

E Case Study Artifact Checklist

To avoid conflating qualitative analysis with unreported experiments, we instantiate each case study only from artifacts produced by the evolution loop. For each of the three cases above, the final appendix should include:

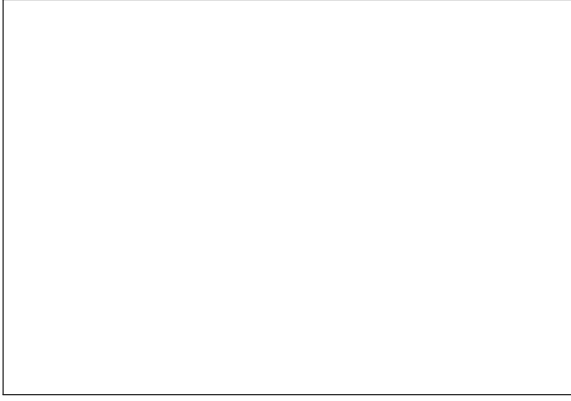


Figure 6: **Placeholder for a concrete before/after case study.** The final version should insert the original image, the failed base answer, the generated skill/tool artifact, the processed visual output if any, and the validated retry answer.

- the triggering failure record: benchmark, example id, question, ground-truth answer, and base-agent answer; 1388 1389
- the generated skill file, quoted or summarized without changing its operative steps;
- the generated tool file when one exists, including the exact operation it performs;
- the visual artifact produced by the tool, such as a marked chart, selected crop, or embedded visual anchor;
- the validation record showing whether the capability was accepted and whether the retry answer changed.

If a case lacks one of these artifacts, we mark the missing field explicitly rather than inferring it from the intended behavior. This keeps the case studies as evidence for the form of the evolved capabilities, while the quantitative claims remain grounded in the tables in Section 4.

F Reproducibility Statement

The code, evolved capability files (learned/ directory), and all evaluation scripts will be released in an anonymised repository upon acceptance. The experiments use publicly available benchmark datasets and the VLM backbones described in Section 4.1. Evolved capability sets (*.skill.md and *.tool.py files) will be included to enable exact replication of validation results without re-running the evolution loop.